

Bibliographic Review, Topic 39

The Blind Spot Paradox

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1 The problem setting

A model deployed on a stream must cope with *concept drift*, a change over time in the relation $P(y | X)$. Moreno-Torres et al. [1] separate it from a shift in $P(X)$ alone, in a taxonomy they build because the literature had been using the same words for different things. The two surveys that frame the field, Gama et al. [2] and Lu et al. [3], describe the same deployed architecture: an adaptive classifier paired with an external detector watching its error rate. Both treat that pair as cooperative, and neither asks whether the classifier might destroy what the detector is trying to measure.

Each block has its own reaction time. The Adaptive Random Forest [4] is a bag of Hoeffding Trees [5], incremental and inert once converged, resampled by Poisson($\lambda=6$) weights, and every tree carries its own ADWIN. ADWIN [6] splits a variable-length window in two and declares a change when the sub-window means diverge past a Hoeffding bound; on a warning the ARF grows a background tree, and on a confirmation it swaps that tree in. The external block is a CUSUM [7] whose Page-Hinkley form accumulates $S_t = \max(0, S_{t-1} + e_t - p_0 - \delta_P)$ and fires past λ . Hinkley [8] supplies the companion result, the asymptotic law of the change point *estimated* from that same statistic. By Wald's equation the accumulation time is $\tau_{\text{det}}^* = \lambda/(\Delta e - \delta_P) = O(1/\Delta e)$.

Both blocks read the same binary error stream, and one erases what the other needs to see. The question matters in finance, where regime switches are abrupt and volatility clusters. ProteuS [9] fits ARMA–GARCH models to four ETFs and interpolates between them, producing synthetic returns with a known break date, the ground truth that real market data cannot supply.

2 What the source paper establishes

Working in River 0.23.0 [10] and instrumenting the ARF's internal swap counter rather than simulating the race with proxy streams, the manuscript establishes three things.

A clock artefact, not an ADWIN flaw. River evaluates ADWIN every c steps, $c = 32$ by default, while the ARF's internal detectors run at $c = 1$. The measured delay is then $\text{ADD} = c - 1 - (\tau^* \bmod c) + \tau_{\text{stat}}$, uniform on $\{\tau_{\text{stat}}, \dots, \tau_{\text{stat}} + c - 1\}$ for an arbitrary drift date. The default pipeline handicaps the external monitor before any statistics are involved.

Starvation, amplified by the Hydra effect. The forest adapts as soon as its fastest tree does, $\tau_{\text{ARF}} = \min_{i \leq M} \tau_i$, and the instrumentation measures that as 4 to 8 times faster than a single tree ($\tau_{\text{ARF}} \approx 18.5 \Delta e^{-0.98}$ against $\tau_{\text{HAT}} \approx 102 \Delta e^{-1.02}$: same exponent, smaller prefactor). The error excess therefore lasts $O(1/\Delta e)$, and so does the CUSUM's accumulation requirement. The two scale together, so a violent shock is repaired faster and leaves the detector no more evidence than a slow one; a Doob maximal inequality over the transient window bounds the crossing probability away from 1. On ProteuS streams, a PHT paired with an ARF at $c_{\text{int}}=1$ detects 0 of 1,080 runs, whereas throttling the forest to $c_{\text{int}}=32$ restores $F_1 \approx 0.6$. The same collapse appears with EDDM and with Streaming Random Patches, which is what makes it a property of the cumulative-evidence family rather than of one detector.

Consequences. Under an assumption of independent trees, $P_{\text{miss}} = 1 - (1 - F(\tau_{\text{det}}^*))^M$, whence a critical ensemble size $M_{\text{crit}} = \lfloor \ln \beta / \ln(1 - F(\tau_{\text{det}}^*)) \rfloor$ beyond which no guarantee survives. At $\Delta e = 0.33$ it evaluates to 0, so the certificate fails even for a single tree. The proposed fix, the Decoupling Principle, requires $c_{\text{ext}} < c_{\text{int}}$ and $\lambda < \lambda^*(\Delta e_{\text{min}})$, and it runs straight into a negative result the authors state plainly: on heteroscedastic streams, decoupling caps λ at 12.4 while false-alarm control demands at least 15, which leaves the admissible set empty. Detection is recovered by changing detector family rather than by tuning. KSWIN [11], a Kolmogorov–Smirnov test between sliding windows, keeps pre-drift data in its reference window and cannot be starved.

3 Where our reading departs from the manuscript

The race is between two random times. The manuscript compares τ_{ARF} against the nominal $\tau_{\text{det}}^* = \lambda/(\Delta e - \delta_P)$, a deterministic quantity, whereas the alarm is itself a stopping time that may never fire. The natural object is the cumulative incidence function of competing risks, $F_1(t) = \Pr(T \leq t, \varepsilon = \text{repair})$ with $T = \min(\tau_{\text{ARF}}, \tau_{\text{det}})$. Fine and Gray [12] model its subdistribution hazard directly, their argument being that the cause-specific hazard carries no usable interpretation in terms of the probability of failing from that cause. Estimation goes through the

competing-risks case of the Aalen–Johansen product-limit estimator [13], not a Kaplan–Meier treating alarms as independent censoring: an alarm structurally prevents observing the repair in the same run, which is the dependence Kaplan–Meier assumes away.

The independence assumption is optimistic, and the bias can be signed. The τ_i enter through an order statistic [14], but the M trees share one stream and are not independent. Conditioning on the stream realization S leaves only the per-tree Poisson weights and feature subspaces, which are i.i.d. across trees. Writing $G_S(s) = P(\tau_1 > s \mid S)$ and applying Jensen’s inequality to the convex map $x \mapsto x^M$ gives $P(\tau_{\text{ARF}} > s) = \mathbb{E}_S[G_S(s)^M] \geq (1 - F(s))^M$. The real repair is slower than the formula predicts, with equality only if the difficulty of adapting does not depend on which stream was drawn, a case that does not arise here since every seed produces a different post-drift noise trajectory. Esary, Proschan and Walkup [15] reach the same place from the other side: their Theorem 5.1 gives $\Pr(\tau_1 > s_1, \dots, \tau_M > s_M) \geq \prod_i \Pr(\tau_i > s_i)$ for associated variables, and the law of total covariance shows that the trees here can only be non-negatively associated. The manuscript concedes the point in a disclaimer; making it quantitative turns M_{crit} into a conservative floor rather than an estimate.

No standard metric exists for what has to be measured. Both surveys enumerate the evaluation toolkit, prequential error, kappa and kappa-temporal statistics, prequential AUC, RAM-hours [2, 3], and neither offers a time-to-recovery measure. That gap is why the group instruments the swap fraction, the smoothed ensemble error and the detector’s internal state together, and why τ_{ARF} , the date of the first swap alone, has to be validated as a proxy for ensemble recovery instead of assumed to be one.

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